Spring Boot REST Steps

Enter Spring boot REST
Understand RESTful web services n it's architecture , which sits on the top of Spring MVC
(Draw a comparison , with existing Spring MVC Flow Diagram)
1.1 Ref : Full Stack Overview.png

1.2 refer to : REST Simplified
1.3 Diag : web app vs web service
1.4 Readme : RestController vs MVC Controller n Annotations.txt
1.5 Develop the backend for Blog management for category resource
(CRUD)

Steps
1. Create spring boot app : using spring boot starter project (choose packaging as JAR)
2. Use same spring boot starters as earlier.
Spring web , Mysql driver , Spring data JPA , Lombok , Spring Dev Tools,validation
2.1 Since Spring boot starter DOES NOT support spring boot 2.x , choose Spring boot 3.2 n JDK 17.
After the whole project is built , it will show you build errors(since currently we don't have JDK 17 with us !)
2.2 Open pom.xml n edit Spring Boot parent version (line no 8)
<version>2.7.18</version>
And Edit Java version
<java.version>11</java.version> (line no 17)

2.3 Force update Maven . Open Maven dependencies n conform that you are actually using spring boot 2.7.x

3. NO additional dependencies for view layer(i.e no jstl n no tomcat-embeded jasper dependencies , in pom.xml

4. Copy application.properties from earlier spring boot project
(Do not add view resolver related properties)

5. Add additional dependencies , in pom.xml
```
<!-- Swagger UI -->
  <dependency>
   <groupId>org.springdoc</groupId>
   <artifactId>springdoc-openapi-ui</artifactId>
   <version>1.7.0</version>
  </dependency>
  <!-- https://mvnrepository.com/artifact/org.modelmapper/modelmapper -->
  <dependency>
   <groupId>org.modelmapper</groupId>
   <artifactId>modelmapper</artifactId>
   <version>3.0.0</version>
  </dependency>
  <!-- https://mvnrepository.com/artifact/commons-io/commons-io -->
  <dependency>
   <groupId>commons-io</groupId>
   <artifactId>commons-io</artifactId>
   <version>2.11.0</version>
  </dependency>
```

OR
If you don’t want to invest time in all of above , simply import "template projects\spring_boot_backend_template" n start with the actual development .

For advanced Java Lab exam , you will be getting this template project. You will have to develop the RestController, Service n DAO layer n Entities on your own , as per the requirements.

6. Write simple RestController to serve simple resource , to confirm.

7. Build the layers in bottoms up manner, for the following objectives.

Objective : Complete backend for the Blog management’s for Category resource

7.1 Continue to use blogs DB .
7.2 Create the packages
Copy entities , value_types ,repository , service

7.3. Add lombok annotations
Refer - "day13_help\Regarding Lombok.txt"
Apply these annotations only for Category resource

7.4 Comment the association between Category 1<--->* Blog Posts.
E-R demo will be discussed later.

Solve

1. Get All Categories
2. Add new Category

3. Delete catgeory details
Better Practice : Instead of a plain string message , wrap it in the java object (ApiResponse) n send it to front end

4. Update category details
4.1 Get existing category details By Id(from the back end) : GET
4.2 Update category details : PUT

8.
Regarding swagger
Refer - "day13_help\Regarding Swagger.txt"

Test it with postman/swagger (Add swagger dependency here)
Steps
1. Add swagger dependency in pom.xml
(Already added in spring boot backend template project)
2. From web browser , access swagger endpoint
http://localhost:8080/swagger-ui/index.html